

Bargaining Theory

Bargaining Theory is a branch of game theory that studies **how two or more individuals, firms, or countries negotiate to divide resources, reach agreements, or resolve conflicts**. It analyzes what outcome is likely when each party has different preferences, bargaining power, and alternatives.

The main question Bargaining Theory answers is:

"If people can benefit from cooperating, how will they divide the benefits?"

Definition

Bargaining Theory is the study of **strategic negotiation**, where participants try to maximize their own payoff while reaching an agreement that is acceptable to everyone involved.

Why is Bargaining Theory Important?

Many real-life decisions involve negotiation rather than competition.

Examples include:

- Salary negotiations
- Business partnerships
- International trade agreements
- Peace treaties
- Labor union negotiations
- Divorce settlements
- Price negotiations
- Contract discussions

Basic Components

1. Players

The individuals or organizations involved in negotiation.

Example:

- Buyer
- Seller

2. Resource to Divide

Something valuable that both players want.

Examples:

- Money
- Profit

- Land
 - Time
 - Business ownership
 - Political power
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3. Bargaining Power

The ability of one player to influence the outcome.

Greater bargaining power usually means receiving a larger share.

4. Reservation Value (Walk-away Point)

The minimum acceptable offer.

If the agreement is worse than this value, the player prefers not to negotiate.

5. BATNA

Best Alternative To a Negotiated Agreement

This is the best option available if negotiations fail.

A stronger BATNA increases bargaining power.

Types of Bargaining

1. Distributive Bargaining

A fixed amount is divided.

One person's gain is another's loss.

Example:

Negotiating the selling price of a used car.

2. Integrative Bargaining

Players cooperate to increase total value before dividing it.

Example:

Two companies collaborate and both earn more profit than working separately.

The Nash Bargaining Solution

One of the most famous bargaining models was developed by John Forbes Nash Jr..

It predicts a fair agreement between two rational players.

The Nash solution assumes:

- Both players behave rationally.
 - Both know each other's alternatives.
 - Agreement benefits both.
 - No player can force the other.
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Nash Bargaining Formula

If

- Player A gains U_A
- Player B gains U_B
- Their disagreement (fallback) payoffs are d_A and d_B

The Nash Bargaining Solution maximizes:

$$(U_A - d_A)(U_B - d_B)$$

This balances the gains above each player's fallback option.

Assumptions

- Rational players
- Complete information
- Voluntary participation
- Players prefer agreement over disagreement
- Utility can be measured

Advantages

- Explains negotiations mathematically.
 - Predicts likely agreements.
 - Helps improve negotiation strategies.
 - Widely applicable in economics, business, law, and politics.
 - Encourages mutually beneficial outcomes.
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Limitations

- Assumes rational behavior.
- Emotions and relationships are difficult to model.
- Real negotiations often involve incomplete information.
- Power imbalances may prevent fair outcomes.

Key Terms

- **Bargaining:** Negotiating to reach an agreement.
- **Reservation Value:** Minimum acceptable outcome.
- **BATNA:** Best alternative if negotiations fail.
- **Bargaining Power:** Ability to influence the agreement.
- **Nash Bargaining Solution:** A mathematical model for predicting a fair division of gains from cooperation.
- **Disagreement Point:** The payoff each player receives if no agreement is reached.